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Digital Material Passport

ID	DMP-METAL-003	Version	1.0.0
Issue Date	2025-05-15	Certificate Type	EN 10204 3.1

Business Transaction

Order		Delivery	
Order ID	PO-34567	Delivery ID	DN-89012
Position	3	Position	1
Date	2025-04-25	Date	2025-05-14
Quantity	500 kg	Quantity	500 kg

Product Information

Product Name	Aluminum Alloy 7075-T6
Batch ID	H-43210-01
Heat Treatment	Solution treated and artificially aged (T6)
Surface Condition	Rolled
Production Date	2025-05-12
Country of Origin	DE

Product Norms

Designation	AMS 4045 (2023)
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Material Designations

System	AA UNS
Designation	7075-T6 A97075

Product Shape

Form	Plate
Length	2000 mm
Width	1000 mm
Thickness	10 mm

Chemical Analysis

Heat Number	H-43210
Melting Process	VAR
Casting Date	2025-05-10
Sample Location	Ladle

Elements

Symbol	Al	Zn	Mg	Cu	Cr	F1
Unit	%	%	%	%	%	
Min	-	5.1	2.1	1.2	0.18	0.18
Max	-	6.1	2.9	2.0	0.28	0.28
Actual	89.7	5.6	2.4	1.5	0.22	0.22

Formula Definitions

F1 = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Tensile Strength	Rm	572 MPa	530		ASTM E8	✓
0.2% Yield Strength	Rp0.2	505 MPa	480		ASTM E8	✓
Elongation	A	11 %	10		ASTM E8	✓

Physical Properties

Property	Symbol	Actual	Target/Min	Maximum	Method	Status
Density	ρ	2.81 g/cm³	2.81		ASTM B311	✓
Coefficient of Thermal - Expansion	α	23.4 10 ⁻⁶ /K	23.5		ASTM E228	✓
Thermal Conductivity	λ	130 W/(m·K)	120		ASTM E1461	✓
Specific Heat Capacity	cp	862 J/(kg·K)	860		ASTM E1269	✓
Electrical Resistivity	ρ _e	0.0538 μΩ·m	-	0.0550	ASTM B193	✓
Poisson's Ratio	ν	0.33	0.33		ASTM E132	✓
Melting Range	Tm	477 - 635 °C	475 - 635		ASTM E1142	✓
Relative Magnetic - Permeability	μ _r	1.00002	-	1.0001	ASTM A342	✓
Surface Roughness	Ra	0.8 μm	-	1.6	ISO 4287	✓
Emissivity	ε	0.09	-	0.11	ASTM E408	✓
Surface Tension	γ	0.875 N/m	0.870		ASTM D971	✓
Diffusion Coefficient	D	2.3E-9 m²/s	2.2E-9		ASTM E1559	✓

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with AMS 4045 and the specified test methods. All results are within the specified limits.

Validated By

Name	Title	Department	Date
Elsa Müller	Materials Engineer	Quality Assurance	2025-05-15